TYLER ALLEN 25allentm@gmail.com | 949-226-2273 | tylerallenengr.github.io | San Clemente, CA | linkedin.com/in/tylerallenengr

SUMMARY

Mechanical Engineering student with extensive hands-on experience in rapid prototyping, CAD design, and systems integration. Passionate about bridging the gap between complex mechanical assemblies and control systems in robotics and autonomous vehicles.

EDUCATION

Purdue University — West Lafayette, IN Aug 2025 – Present

- Mechanical Engineering Major, May 2029 expected graduation

San Juan Hills High School — GPA 4.13 (Valedictorian status), SAT Score 1480 Aug 2021 – May 2025

ENGINEERING EXPERIENCE

Aerial Robotics Team, Purdue University Sept 2025 – Present
RND Mechanisms Propulsion Subsystem Lead

- End-to-End Mechatronic System Design: Sole designer of the tilt-rotor mechanism for both forward motors and static rear motor mount, using parametric SolidWorks CAD with tight tolerances — prioritizing minimal moving parts and reduced failure points to meet spatial and mass constraints.
- Structural Validation & Avionics Integration: Performed static load testing on all designed components to verify structural integrity under operational loads; collaborated with the avionics team to diagnose and resolve system-level hardware issues.

FRC Robotics Team 5199, Capistrano Unified School District
Vice President Aug 2024 – May 2025

- Led 48-person robotics team in CAD design and strategy, advancing to FRC World Championship and earning team MVP
- Specialized in mechanical systems and expert level Onshape CAD design
- Won the FRC 'Excellence in Engineering Award,' 'Quality Award,' and a Regional Championship
- Collaborated directly with the programming sub-team to ensure precise integration between mechanical assemblies, sensors, and robotic control systems for reliable physical operation.

Department Lead, Business Operations, Fundraising & Outreach Aug 2022 – May 2025

- Managed department of eight students to raise over $200,000 through outreach and grant campaigns
- Organized outreach events at elementary and middle schools, and Women in STEM events

COSMOS Summer Program, University of California, San Diego, CA Aug 2024 – Aug 2024
Participant

- Built autonomous data collection USV with a team of four
- Integrated a sensor array with an Arduino Uno that used LoRa radio to transmit environmental data to a base station database in real time
- Configured ArduPilot flight controller telemetry and tuned autonomous waypoint navigation to reliably execute data collection missions.

Solo Project, Custom FPV Quadcopter July 2025 – Aug 2025
Builder

- Engineered 5-inch propellor drone, integrated ESCs and power distribution, and performed PID tuning and sensor calibration to ensure stable flight dynamics.

WORK EXPERIENCE

Oceanside Gym Jun 2025 – Aug 2025
Front Desk Manager

- Managed daily gym operations independently- including opening/closing, member tours, and payment processing.

SKILLS

- **Technical**: expert-level Onshape CAD, proficient in SolidWorks, proficient in data analysis using Python, Excel, and MATLAB.
- **Equipment Experience**: Bambu Labs 3D Printers, bandsaw, chop saw, table saw, jigsaw, drill press, rotary sander, Manual and Haas CNC mill.
- **Systems Integration**: ArduPilot, Flight Controller Configuration, Sensor Integration (IMU, GPS, etc.), PID Tuning, Soldering, Molex Crimping, Arduino.